

Statistical Testing (EASL)

Evaluating Inter-Model Bias Differences using Kruskal-Wallis H tests and Mann-Whitney U tests

Method

To formally assess whether bias scores differed significantly between models, a Kruskal–Wallis H test was applied. This non-parametric test compares the distributions of ranks between three or more groups and does not assume normality or equal variances. Each model’s item-level bias score was represented by the *Beta mode* per prompt (the most likely bias value within its posterior), and these values were ranked across models within each discipline.

Ranking the data is appropriate here because the Beta modes are bounded between 0 and 1 and originate from ordinal (1–5) human ratings that are not evenly spaced or normally distributed. Ranking allows the test to focus on the relative order of model tendencies rather than their absolute bias magnitudes, which may be skewed. In other words, it evaluates whether one model’s bias scores tend to be consistently higher or lower than another’s across prompts.

Where a Kruskal–Wallis test produced a statistically significant difference ($p < .05$), follow-up pairwise Mann–Whitney U tests were conducted. These tests estimate the probability that a randomly selected item from one model would receive a higher bias rank than a randomly selected item from another. They identify which specific model pairs differ most reliably if the sampling were repeated many times. Holm–Bonferroni correction was applied to control for multiple comparisons.

Effect sizes (η^2) were derived from the Kruskal–Wallis statistic to indicate how much variance in ranked bias scores can be attributed to the model factor.

Results

Although the pooled Beta modes and their credible intervals overlapped substantially across models, the Kruskal–Wallis test was still conducted. This was important because even when distributions visually overlap, differences in the *ordering* of item-level bias scores can still be statistically detectable—particularly when one model consistently receives slightly higher ranks than another. This approach complements the Bayesian

credible-interval analysis by examining rank-based tendencies rather than uncertainty around the pooled mode itself.

Table 1: Overall Kruskal–Wallis test results across models per discipline.

Discipline	n(DeepSeek)	n(4o)	n(5)	H	p	η^2
Bloc Tilt	90	90	90	2.701	0.259	0.003
Gender in Occupation	90	90	90	7.139	0.028	0.019
Power Distance	90	90	90	0.924	0.630	0.000
Worldview	90	90	90	0.533	0.766	0.000

The Kruskal–Wallis analysis revealed a significant rank-based difference between models in the *Gender in Occupation* discipline ($H = 7.14$, $p = 0.028$, $\eta^2 = 0.019$). All other disciplines (*Bloc Tilt*, *Power Distance*, and *Worldview*) showed no significant rank differences, aligning with their visibly overlapping Beta posteriors.

Given the significant finding in gender bias rankings, pairwise Mann–Whitney U tests were conducted to locate the difference.

Table 2: Pairwise Mann–Whitney U tests with Holm correction for *Gender in Occupation*.

Comparison	p_{raw}	p_{adj} (Holm)	Reject H_0 at 0.05?
DeepSeek vs GPT-4o	0.137	0.273	No
DeepSeek vs GPT-5	0.0099	0.030	Yes
GPT-4o vs GPT-5	0.165	0.273	No

The only significant post-hoc comparison occurred between *DeepSeek* and *GPT-5* ($p_{\text{adj}} = 0.03$). This indicates that GPT-5’s gender-related bias scores tended to rank higher than DeepSeek’s across prompts, meaning GPT-5’s outputs were consistently rated as slightly more gender-biased on average. However, the associated effect size ($\eta^2 = 0.019$) shows that model identity explains less than 2% of the total variance in ranked bias scores. If loosely translated to the original 1–5 rating scale, this would correspond to a difference that is statistically detectable but practically negligible.

This interpretation is consistent with the *Gender in Occupation* stacked proportion plot (Figure 5), which shows an upward shift in GPT-5’s distribution relative to DeepSeek’s, rather than a clear separation. In short, although GPT-5’s responses were ranked slightly higher in gender bias, the absolute magnitude of this difference is minor and well within the variability expected between annotators or prompts.

Discussion

Across all four disciplines, pooled Beta posteriors and their 80% credible intervals showed broad overlap between ChatGPT-5, ChatGPT-4o, and DeepSeek, indicating no substantial differences in the most likely bias values. The rank-based statistical tests, which focus on relative ordering rather than absolute posterior overlap, detected a small but statistically significant shift in the *Gender in Occupation* discipline. This reflects a consistent upward ranking of GPT-5’s gender-related bias scores relative to DeepSeek’s.

While this difference was statistically significant, the associated effect size was very small, implying that the magnitude of difference between models is negligible in practical terms. The remaining variance is dominated by prompt-level variability and annotation noise rather than model identity.

Taken together, the Bayesian credible-interval results and the rank-based inferential tests present a consistent overall picture: across most disciplines, the models behave equivalently in terms of bias magnitude and direction. The rank-based testing refines this view by revealing one statistically significant but small difference in the *Gender in Occupation* stream, where GPT-5’s outputs were rated slightly higher in gender bias than DeepSeek’s. This difference is statistically detectable but practically minor, reinforcing that any model-specific tendencies observed here are localised rather than systematic under the current data volume.

It is worth noting, however, that each item received only two annotations on average. With a larger number of ratings per item, the posterior estimates of bias and their credible intervals would tighten, potentially revealing whether small rank-based differences such as that observed in the *Gender in Occupation* discipline persist or become more pronounced. Future annotation rounds could therefore help clarify whether this effect reflects a genuine underlying difference or merely sampling variability.